Cuts

Cuts describe which leaves a subgraph may stand on for techmapping

The idea

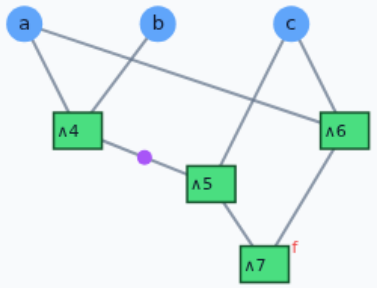
- PI cut={self}
- AND cuts = unions of fanin cuts with size ≤ k, plus trivial {self}

K=3

CUTS

K=3

Step 1 / 5 · k · module02-05-cut-enumeration



Explanation

Toy uses cut_k=3.

- k-feasible

k=3

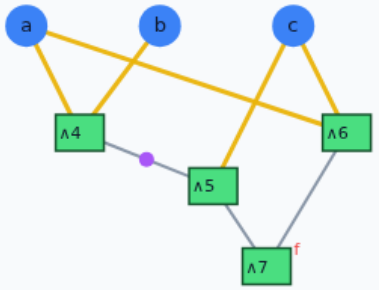
K=3

PI cuts

CUTS

PI cuts

Step 2 / 5 · pi · module02-05-cut-enumeration



Explanation

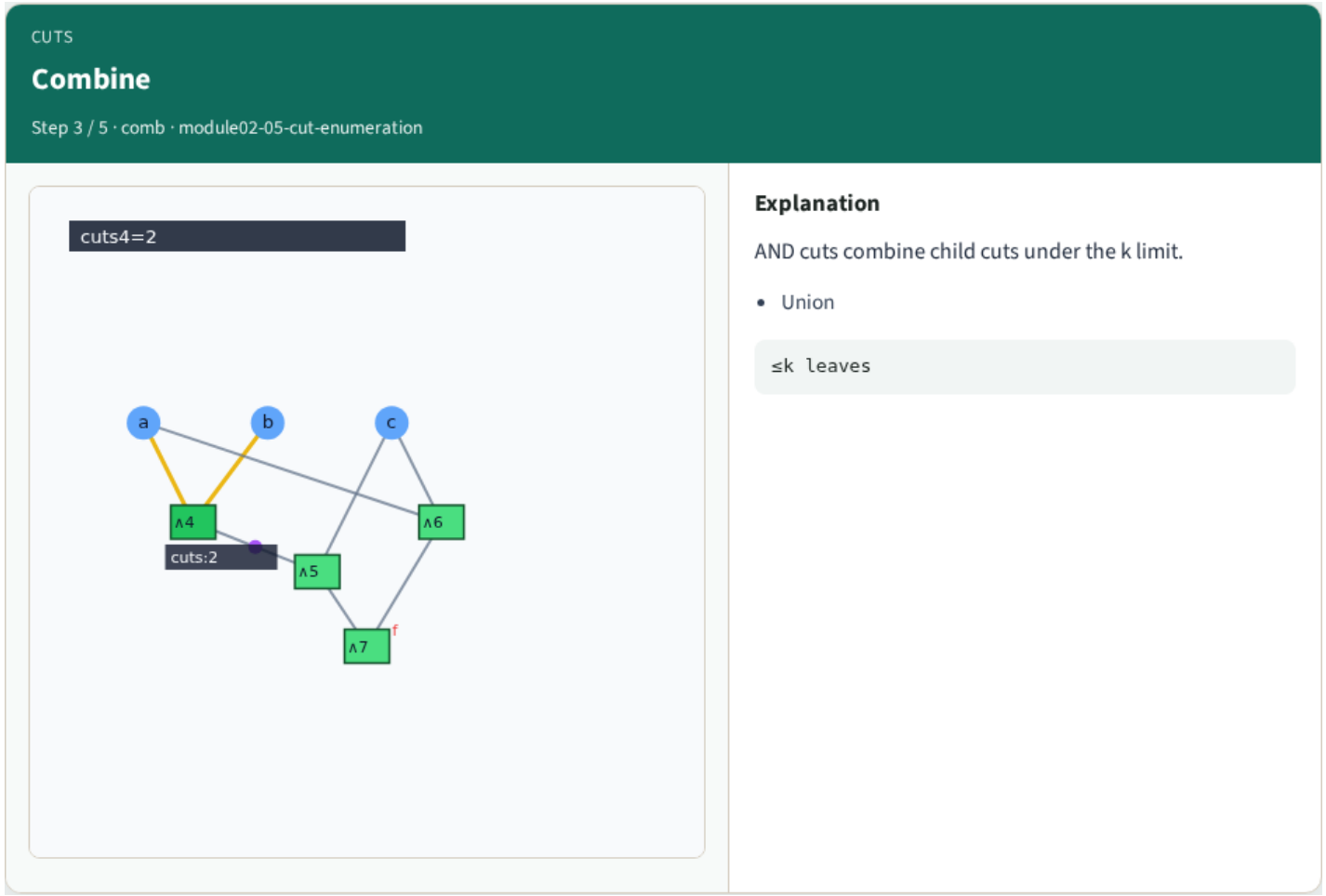
Each PI has a single cut.

- {self}

```
3 PIs
```

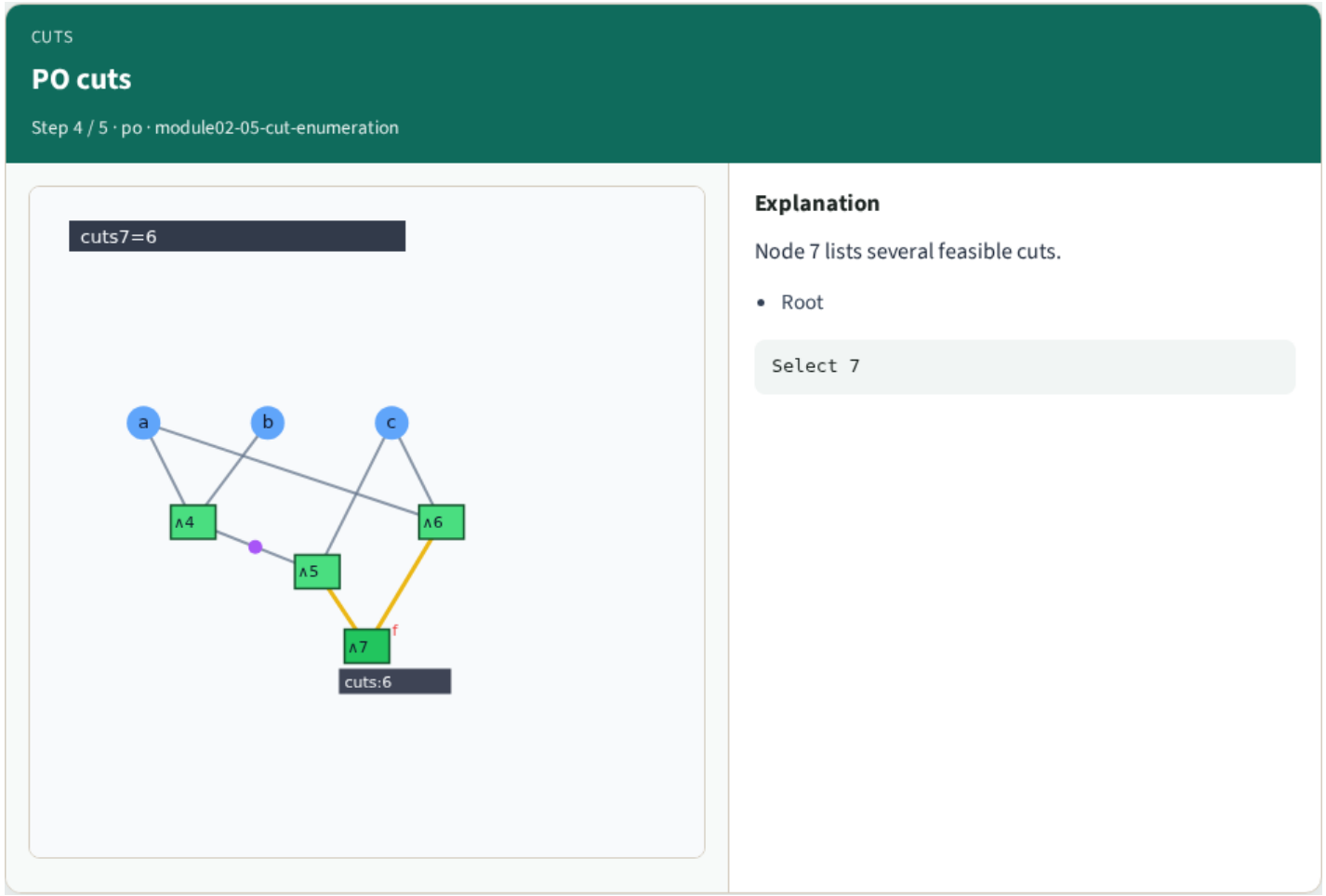
PI cuts

Combine



Combine

PO cuts



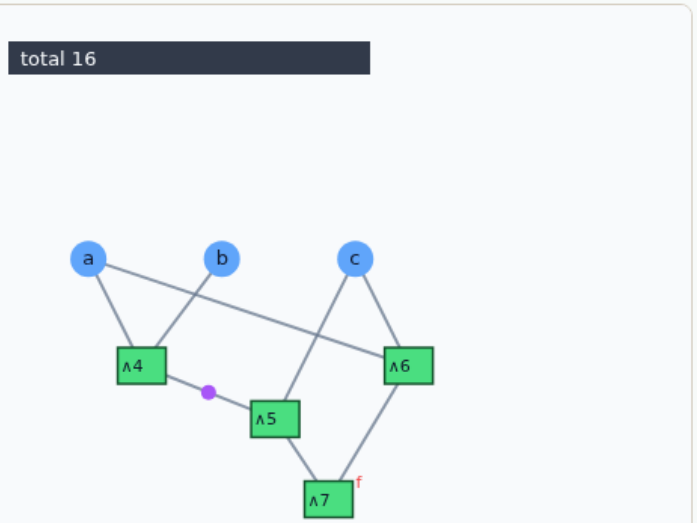
PO cuts

Stats

CUTS

Stats

Step 5 / 5 · stats · module02-05-cut-enumeration



Explanation

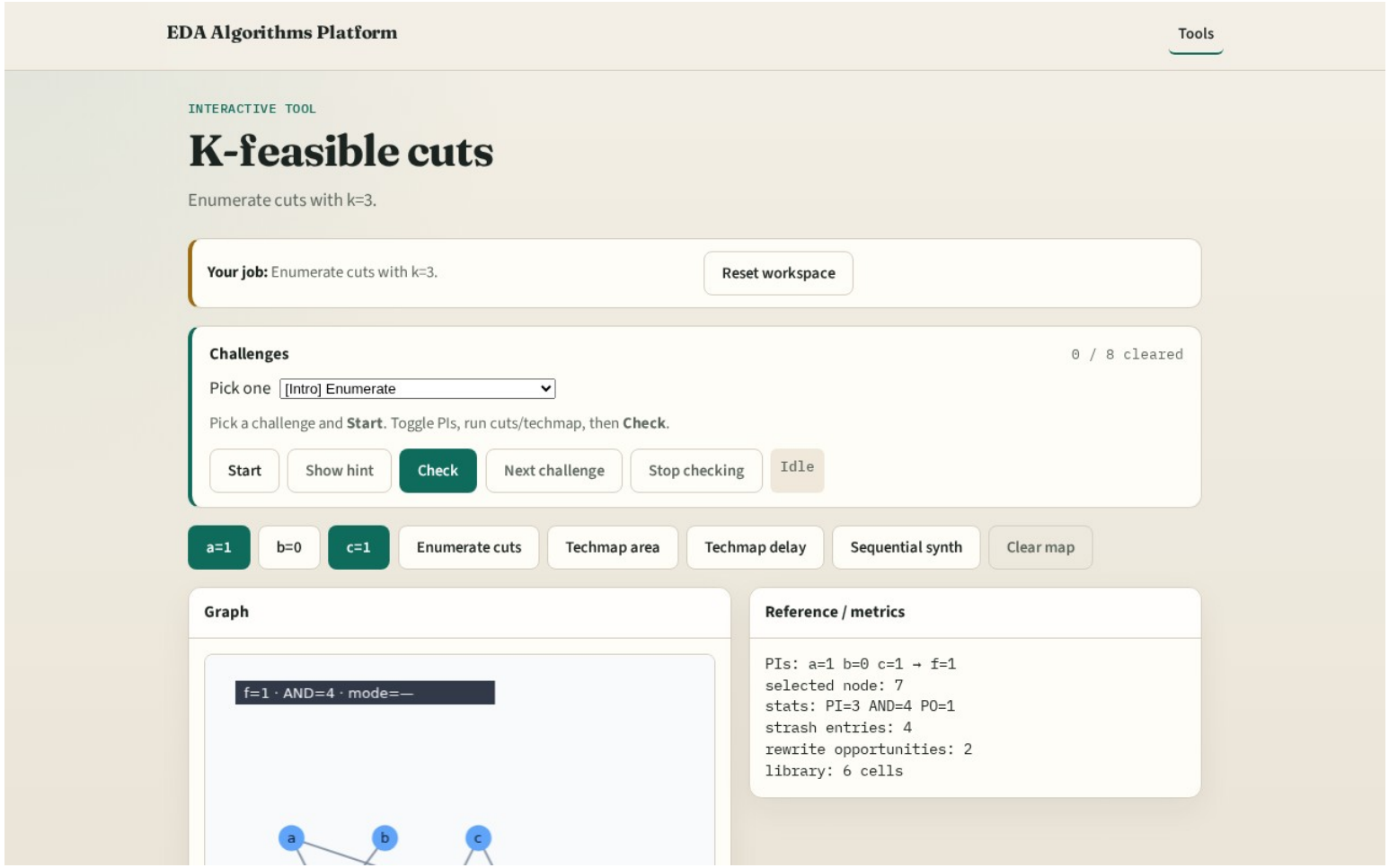
total_cuts grows with graph depth.

- cutStats

```
total=16
```

Stats

Browser lab track



Browser lab starter

Implement track

- Call enumerate_cuts(k=3); assert all |cut|≤3

Pitfalls

- Allowing cuts larger than k
- Forgetting the trivial cut

Your turn

- Next: the cell library